Year 12 Maths - Objects in contact

Similar questions - Pulleys and connected particles

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 Particles P and Q, of masses 4 kg and 6 kg, are attached to the ends of a light inextensible string that passes over a smooth fixed pulley. The system is held at rest with the string taut, Q at a height of 1.5 m above the ground, and then released.

(a) Find the acceleration of the particles and the tension in the string. **[5]**

(b) Find the speed of Q when it hits the ground. **[2]**

(c) Q does not rebound. Find the distance P moves while the string is slack, and the time for which the string is slack. **[5]**

(d) State one modelling assumption that you have used about the string, and say how it affects your answer to part **(a)**. **[1]**

2 A train consists of an engine of mass 12 000 kg pulling a truck of mass 8000 kg along a straight horizontal track. The resistances to motion are 1500 N on the engine and 1000 N on the truck. The engine produces a driving force of 7000 N.

(a) Find the acceleration of the train and the tension in the coupling. **[5]**

(b) The train passes a signal with speed 5 $\mathrm{m\,s^{-1}}$. Find its speed and the distance travelled 80 seconds later. **[3]**

(c) At that moment the coupling breaks. Find the distance the truck travels before coming to rest, assuming the resistance on the truck remains 1000 N. **[3]**

Question 1

(a) Q: $6g - T = 6a$. P: $T - 4g = 4a$. Adding: $2g = 10a$, so $a = 0.2g = 1.96\ \text{m s}^{-2}$. $T = 4(g + a) = 47.04$ N

(b) $v^2 = 2as = 2(1.96)(1.5) = 5.88$, so $v = 2.42\ \text{m s}^{-1}$

(c) After Q stops, P is a projectile moving up at $2.42\ \text{m s}^{-1}$ under gravity alone. It rises $\dfrac{v^2}{2g} = \dfrac{5.88}{19.6} = 0.3$ m and then falls 0.3 m back, when the string becomes taut again: distance 0.6 m. Time slack $= \dfrac{2v}{g} = \dfrac{2(2.425)}{9.8} = 0.495$ s

(d) The string is inextensible, so P and Q have the same acceleration magnitude (the tension is the same throughout as the string is light and the pulley smooth). If the string could stretch, the particles would have different accelerations and the tension would not equal 47.04 N

Question 2

(a) Whole train: $7000 - 2500 = 20\,000a$, so $a = 0.225\ \text{m s}^{-2}$. Truck: $T - 1000 = 8000a$, so $T = 2800$ N

(b) $v = 5 + 0.225 \times 80 = 23\ \text{m s}^{-1}$. $s = 5(80) + \frac{1}{2}(0.225)(80)^2 = 1120$ m

(c) Deceleration of the truck $= \dfrac{1000}{8000} = 0.125\ \text{m s}^{-2}$. $s = \dfrac{23^2}{2(0.125)} = 2116$ m